DE–6697 **11**

DISTANCE EDUCATION

P.G.D.C.A. (S) DEGREE EXAMINATION, DECEMBER 2013

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Explain the various elements of a computer and communication systems.

2. Explain the developments in communication technology.

3. Elaborate on Ethics and Intellectual property rights.

4. Explain the various types of application software.

5. Explain the telephone related communication services.

6. Explain the features of any four secondary storage devices.

7. Explain the different types of database organizations.

8. Explain the various phases in System Analysis and Design.

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DE–6698 **12**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

First Semester

DIGITAL COMPUTER ORGANISATION

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Convert :

11110000_2 to Octal equivalent and decimal

4433_8 to Decimal equivalent and binary equivalent.

(b) Convert Hexadecimal to decimal, binary and octal

(i) $AB6_{16}$

(ii) $3A6_{16}$.

2. (a) Write short notes on :

(i) Sum of product.

(ii) Product of sums with examples.

(b) Simplify the Boolean expression using Boolean relation and draw the circuit.

$Y = \overline{A}BCD + \overline{A}BC + A\overline{B}CD + ABC$.

3. (a) Simplify the following using K-map

$$\sum m\,(0, 1, 2, 4, 6, 9, 11)$$

(b) Find out the solution for the given expression

$$f(m) = (\overline{A}\,\overline{B}\,\overline{C}\,\overline{D} + ABC\overline{D} + A\,B\,\overline{C}\,\overline{D} + A\overline{B}CD + \overline{A}BC\overline{D}),$$

$d(\overline{A}BCD + ABCD)$ using K-map.

4. (a) Explain encoder and decoder with suitable example.

(b) Write short notes on :

(i) Full-adder

(ii) Full-subractor.

5. (a) Explain the functions of JK Flip Flops with neat diagram and truth table.

(b) With sketch, explain the operation of master-slave flip flop.

6. (a) Explain RAM and ROM and list out the advantages of ROM and RAM.

(b) What is parallel adder? What are its advantages over full adder?

7. (a) Explain about Magnetic Tape Memory with suitable diagram.

(b) Write short note on Direct Memory Access.

8. (a) Explain in detail Interrupt with an example.

(b) Draw a multiplexer using only NAND gates which selects from four inputs I_0 to I_3 using two select inputs S_0 and S_1.

DE–6699 **13**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Academic year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain any four data types in C with examples. (10)

   (b) Explain the arithmetic and logical operators in C. (10)

2. (a) Explain the use of switch statement with an example. (10)

   (b) Write a C program to find factorial of a given number using while statement. (10)

3. (a) Explain the method of declaring and intializing one dimensional array. (10)

   (b) Write a C program to find largest of n given numbers. (10)

4. (a) Explain the use of macros with an example. (10)

   (b) Explain the method of defining and using a structure with an example. (10)

5. (a) Explain the functions used for reading the content of a file. (10)

   (b) Write a C program to swap two integers using pointers. (10)

6. (a) Explain any two operations on queue. (10)

   (b) Explain the operations on stack. (10)

7. (a) Explain the difference between singly linked list and doubly linked list. (10)

   (b) Explain the representation of binary tree. (10)

8. (a) Explain the method of inserting an element into the linked list. (10)

   (b) Discuss any one of the applications of tree. (10)

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DE–6700 **14**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

First Semester

OFFICE AUTOMATION

(2005 to 2010 Academic year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. (a) Discuss windows desktop in detail. (10)

   (b) Explain the procedure for creating, opening and saving files in MS-office. (10)

2. (a) Explain the following features in word.

   (i) Changing text font, style and size (2.5)

   (ii) Align text (2.5)

   (iii) Indent text (2.5)

   (iv) Paragraph spacing (2.5)

   (b) How to create tables in MS-Word? Explain the ways of selecting and moving text within a table. (10)

3. (a) Discuss mail merge with example. (10)

   (b) Explain the following features in excel.

   (i) Inserting data and time (4)

   (ii) Find and replace data (3)

   (iii) Inserting header and footer. (3)

4. (a) Describe the ways of creating different types of charts in excel. (10)

   (b) Explain sorting and filtering procedure in excel. (10)

5. (a) How will you create and manipulate slides in power point? (10)

   (b) Discuss the procedure for adding slide transition effect and animation effect in a power point presentation. (10)

6. (a) Explain the following features in power point

   (i) Setting background (3)

   (ii) Using design template (4)

   (iii) Auto layout. (3)

   (b) Discuss the following operations in MS access with an example. (10)

   (i) insert a record

   (ii) delete a field

   (iii) sorting the records

   (iv) creating primary key.

7. (a) What are queries in access? Explain the query design environment. (10)

   (b) How to create form in MS Access and what are the uses of forms? (10)

8. (a) Explain the following operations in word with an example.

   (i) cut a text and place it in another position (4)

   (ii) adding a page number at the center of the page (3)

   (iii) set the margins. (3)

   (b) Explain steps to create a report in MS-Access. (10)

DE–6701 **21**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Discuss the basic concepts of object oriented programming in detail. (14)

   (b) Give a note on object oriented languages. (6)

2. Explain the basic and user defined data types with examples. (20)

3. With neat diagram, syntax and examples explain the control structures used in C++. (20)

4. (a) Explain the concepts of function proto typing and function overloading with syntax and examples. (12)

   (b) Develop a C++ program to overload a functions "Volume( )" at least two times. (8)

5. (a) Explain how do member functions are defined. (12)

   (b) Write a C++ program with class implementation and member functions. (8)

6. With a C++ program, explain the multiple constructors in a class. (20)

7. Illustrate and explain the overloading procedure of binary operators. (20)

8. Write short notes on :

   (a) Opening and closing a file. (10)

   (b) File pointers and their manipulations. (10)

DE–6702 **22**

DISTANCE EDUCATION

P.G.D.C.A. (S) DIPLOMA EXAMINATION, DECEMBER 2013.

Second Semester

SYSTEM ANALYSIS AND DESIGN

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

All questions carry equal marks.

Answer any FIVE questions.

($5 \times 20 = 100$)

1. Explain the characteristics and elements of a system.

2. Explain the multifaceted role of a systems analyst.

3. Explain the steps in feasibility analysis.

4. Explain the various cost/benefit categories with examples.

5. Explain the various stages in Input/Output design.

6. Explain the different types of file organizations.

7. Explain in detail system testing.

8. Explain the major phases in hardware and software selection.

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DE–6703 **23**

DISTANCE EDUCATION

P.G.D.C.A.(S) DEGREE EXAMINATION, DECEMBER 2013.

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 Academic year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

All questions carry equal marks.

1. (a) Explain about the functions of different toolbars in VB. (10)

   (b) Discuss about overview of developing a VB application in detail. (10)

2. (a) Write a VB program to find the factorial of a given number. (10)

   (b) Explain the various data types available in VB. (10)

3. (a) Discuss about any five string functions in VB. (10)

   (b) Explain about events and methods in detail. (10)

4. (a) Discuss about passing an array to functions and procedures with an example. (10)

   (b) Explain about input box, picture box and Richtext box with example. (10)

5. (a) Explain about the various built-in objects in VB. (10)

   (b) Discuss about menu editor with an example program. (10)

6. (a) Write a VB program to generate a simple pay bill for an employee in the company. (10)

   (b) Discuss about geometric objects and animation with an example. (10)

7. (a) Explain about sequential and random access files in detail. (10)

   (b) Write a VB program to develop a form to select any type of file from any directory in any drive using directory box, drive box, file list. (10)

8. (a) Explain how to built Active X control in detail. (10)

   (b) Discuss the creation and use of MDI form. (10)

DE–6704 **24**

DISTANCE EDUCATION

P.G.D.C.A.(S) EXAMINATION, DECEMBER 2013.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

1. (a) With neat diagram, explain the working principle of a Cathode Ray Tube. (12)

   (b) Narrate the functions of various hard copy output devices. (8)

2. Explain the Bresenham's line drawing algorithm in detail. (20)

3. List and explain the basic transformation principles with matrix representations. (20)

4. With procedure, explain the Cohen – Sutherland line clipping algorithm. (20)

5. Discuss on three dimensional rotation transformation in detail. (20)

6. Narrate the concepts of three dimensional viewing with neat diagrams. (20)

7. Explain the depth – buffer algorithm in detail. (20)

8. (a) Describe the concept of command language with its issues. (12)

   (b) Give a note on Information display. (8)

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DE–7603 **15**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

First Semester

COMPUTER LAB — I (DATA STRUCTURES USING C)

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by LOT system.

(Programs must be written in C)

1. (a) Write a program to find the roots of a quadratic equation $ax^2 + bx + c = 0$.

   (b) Write a program to find a word in palindrome or not. (Rg. ADA is palindrome)

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2. (a) Write a program for matrix manipulation (addition, subtraction and multiplication).

   (b) Write a program to implement 'push' and 'pop' operations on stack.

3. (a) Write a program to sort '*n*' numbers in ascending order.

   (b) Write a program to evaluate the given arithmetic expression using stack.

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4. (a) Write a menu driven program to find an element using linear and binary search methods.

   (b) Demonstrate the various string manipulations.

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5. (a) Write a program to print the given three digit number into words.

   (eg. 123 One Two Three)

   (b) Explain the various operations on Queue.

---------- Cut here ----------

6. (a) Write a program to find the maximum and minimum value in an array of '*n*' numbers.

   (b) Write a program to find nCr and nPr values using functions.

$$nPr = \frac{n!}{(n-r)!} \text{ and } nCr = \frac{nPr}{r!} .$$

---------- Cut here ----------

7. (a) Write a program to implement insert and delete operations on Queue.

   (b) Write a program to copy contents of one file to another file.

8. (a) Write a program to count number of characters and numbers of lines in a file.

(b) Write a program to find the largest element in an array and its position.

DE–7604 **16**

DISTANCE EDUCATION

P.G.D.C.A. (Sem) EXAMINATION, DECEMBER 2013.

First Semester

LAB II — MS Office and Internet

MS OFFICE AND INTERNET

(2005 to 2010 Academic year)

Time : Three hours Maximum : 100 marks

Break - up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging / Execution : 35

Result : 10

Total : 100

ONE question should be selected by the Examiner and given to the candidates by lot system.

1. (a) Prepare an invitation for a function to be conducted in your institution. Use different text orientation and pictures to make it attractive.

   (b) Create a HTML page that plays video and audio file.

2. (a) Create a lecture notes and provide audio explanation with the help of sound file in MS-WORD.

(b) Create a HTML page that receives employee name, age, street, city, pin data. Receive male /female and married/ unmarried details through option buttons. Receive hobbies through check boxes.

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3. (a) Create a worksheet in excel for preparing electricity bill with customer no, customer type ('D' for Domestic, 'I' for Industrial type), previous reading, current reading, units and amount per unit. Find total units for each customer and calculate the bill amount.

(b) Create a HTML page that divides the screen space into three frames (one vertical and two horizontal of equal size). Display three different files in them with scrolling titles relevant to their contents.

---------- Cut here ----------

4. (a) Create inventory table with fields item no, name, quantity on hand, reorder level etc,

(i) Create a form in customer format

(ii) Create a query REORDER to show all the items.

(iii) Create a macro that executes REORDER query automatically.

(b) Create a HTML page to highlight all the features of MS office packages in another page, when the user clicks any one feature name of the office packages.

5. (a) Enter 10 sets of values for X and Y in two columns of excel work sheet and find mean, median and standard deviation. Also draw a bar chart for these data.

(b) Create a HTML page to display 10 counses and their highlighted features is 2 different frames using link.

---------- Cut here ----------

6. (a) Create a form letter that informs the customer about the date of maturity of a deposit amount in a commercial bank and request the customer for renewal. Mail merge it with an access database containing all the customer data. Prepare letters for customers whose due date falls in a specific range of dates.

(b) Create HTML page that divides the screen into two frames (Vertically / Horizontally). Display cricket players in one frame and their highlighted performance in another frame when the user clicks over any one of the names of the player.

---------- Cut here ----------

7. (a) Draw Line, Bar, Pie charts for the data given below

ABC company limited

Year	Sales	Cost	Profit
1991	1000	400	600

Add data for 10 years 1991 to 2000. Provide titles, legends, grids and data labels.

(b) Create a HTML page for inviting your friends for the function at your home. Add appropriate picture and scrolling text.

8. (a) Create two tables SALES 1/ SALES 2 and join them to produce a third table SALES 3 in MS-ACCESS.

(b) Create a HTML page that plays video and audio files.

DE–7605 **25**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

Second Semester

COMPUTER LAB III – GRAPHICS USING C++

(2005-2010 Academic Year)

Time : Three hours Maximum : 100 marks

1. One question should be selected by the Examiner and given to the candidates by lot system.
2. Each question carries 100 marks.

Break up marks

Record Note Book	: 5
Algorithm	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. Develop a C++ program to draw a straight line using Bresenham's line drawing algorithm.

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2. Develop a C++ program to draw any one three dimensional object and perform animation on it.

---------- Cut here ----------

3. Write a C++ program to implement the Bresenham's algorithm for circle drawing.

4. Develop a C++ program to draw ball. Perform animation on it. Also make the ball to move from left to write and vice-versa.

---------- Cut here ----------

5. Develop a C++ program to show graphical transformations (all three basic types) of a two dimensional object.

---------- Cut here ----------

6. Write a C++ program to draw a non-linear object "Ellipse".

---------- Cut here ----------

7. Develop a C++ program to implement polygon clipping algorithm.

---------- Cut here ----------

8. Write a C++ program to draw a circle. Also divide the circle into 4 quadrant and fill each quadrant with color.

---------- Cut here ----------

9. Develop a C++ program to draw non-linear objects arcs and curves.

---------- Cut here ----------

10. Develop a C++ program to draw a triangle with equal sides. Also apply basic transformation principles to the drawn two dimensional object "Triangle".

DE–7606 **26**

DISTANCE EDUCATION

P.G.D.C.A.(S). EXAMINATION, DECEMBER 2013.

Second Semester

Computer LAB IV – VISUAL BASIC

(2005 to 2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

One question should be selected by the examiner and given to the candidate by lot system.

Break up marks

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. Develop a VB project to read a number in a text box, to add the individual digits till it is reduced to a single digit and to display it along with messages.

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2. Write a VB project to read a month and year in two text boxes and to display the number of days in that month in a text box.

3. Develop a VB project to perform addition, deletion and modefication operations with the employee records available in a database.

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4. Develop a VB project to read names one–by–one from textbox and to write them in a file. Also read the contents of to display the total number of characters in that file.

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5. Write a VB project to simulate a simple calculator with four operations +, –, * and % along with memory feature.

---------- Cut here ----------

6. Write a VB project to create a screen saver that displays a list of pictures with one second delay between pictures.

---------- Cut here ----------

7. Develop a VB project to create a NOTEPAD title editor with FILE and EDIT options and also display floating menu wherever necessary.

---------- Cut here ----------

8. Write a VB project using ActiveX control to create a text box that accepts only numeric value. Provide the properties for the text box : Back color, Fire color and Text.

---------- Cut here ----------

9. Write a VB project using ActiveX DLL or EXE add a clan module that would perform the following :

   (a) Test whether the given string in Palindrome or not.

   (b) Test whether the given number is odd.

   (c) To find the factorial of a given number.

10. Write a VB project to create a List box and do the following operations :

(a) Add entries.

(b) Delete entries.

(c) Count the number of entries.

(d) Clear the list box.